MODULE 16 / LESSON 01

Evaluate RAG as Separate Components

Evaluate at four boundaries:

DSACADEMY

CORE CONCEPT

An Evaluation Dataset Represents Real Questions

A RAG evaluation example should record: Include normal, rare, multi-chunk, exact-term, paraphrase, unsupported, and access-control cases.

question;	**user or tenant context;**	**expected relevant chunk IDs;**
reference answer or required facts when available;	**unsupported or denied expectation;**	**importance and category;**

RUN IT AND INSPECT THE RESULT

Retrieval Precision and Recall Measure Different Goals

```
expected = {"A", "B", "C"}
retrieved = ["A", "B", "X", "Y"]
hits = expected.intersection(retrieved)
print(len(hits)/len(retrieved), len(hits)/len(expected))
```

OUTPUT

```
0.5 0.6666666666666666
```

Precision = 2/4 = 0.50 Recall = 2/3 = 0.667

REASONING SEQUENCE

Top-k and Filters Change the Candidate Set

Frame	Transform	Validate	Explain
tenant or user access;	document type;	publication status;	date;
1	2	3	4

At k=3: precision 0.67, recall 0.50 At k=8: precision 0.38, recall 1.00 Higher k recovers all evidence but adds more irrelevant chunks. Reranking or better queries may improve the tradeoff.

CORE CONCEPT

Query Rewriting Can Improve Retrieval

A query rewrite transforms a user's question into a search query. Possible methods: Rewriting can improve recall but may change intent. Preserve the original question and record every rewritten query.

- **resolve conversation references;**
- **expand abbreviations;**
- **generate alternate wording;**
- **split a multi-part question;**
- **add domain identifiers.**

RUN IT AND INSPECT THE RESULT

Hybrid Search Combines Exact and Semantic Retrieval

```
def rrf_score(ranks, c=60):
    return sum(1 / (c + rank) for rank in ranks)
```

OUTPUT

A fusion score based on rank positions rather than incompatible raw scores.

Documents found by both methods gain combined evidence.

CORE CONCEPT

Reranking Applies a Stronger Model to Fewer Candidates

A reranker scores query-document pairs using a model more expensive than first-stage retrieval. Pipeline: Reranking can improve ordering but adds latency and cost. It cannot recover a relevant chunk missing from the candidate set.

retrieve a broader candidate set quickly;

rerank those candidates;

keep the best few for context.

REASONING SEQUENCE

Answer Quality Has Several Dimensions

Frame

Correctness: agreement with a reference answer or verified facts

1

Transform

Relevance: whether the response addresses the question

2

Validate

Groundedness: whether claims are supported by supplied context

3

Explain

Citation correctness: whether cited passages support nearby claims

4

Question asks refund period and exclusions. Answer gives 14 days with support but omits annual-plan exclusions. It may be grounded and partly correct, but incomplete.

RUN IT AND INSPECT THE RESULT

Model Judges Need Human Calibration

```
agreement = sum(j == h for j, h in zip(judge_labels, human_labels))
print(agreement / len(human_labels))
```

OUTPUT

```
0.84
```

The 8 disagreements should be grouped by cause before trusting the judge at scale.

QUALITY GATE

Failure Analysis Produces the Next Experiment

- [x] source and parsing status;
- [x] expected and retrieved chunks;
- [x] ranks and filters;
- [x] rewrite;

QUALITY GATE

An Experiment Report Includes Quality, Latency, and Cost

- [x] corpus and ingestion version;
- [x] embedding model and index settings;
- [x] retriever, filters, k, rewrite, and reranker;
- [x] prompt and generation model;

GUIDED LAB

Guided Lab: Improve RAG with Evidence

01

label expected chunk IDs and required facts

02

calculate retrieval precision and recall

03

compare at least three k values

04

test metadata filters